Retro Game Jam GDD Template

Game Title:

Galanought Infinite

Team Name:

Graves Squad

Team Members:

List all members and their student IDs:

- Cameron “Graves” Gillespie (82137761)
- Sam “SombreSole” Smith (82723867)
- Darvesh ”AimedEquation” Dhillon (93792406)

Core Concept

Briefly describe the game.

- What classic retro game are you taking inspiration from? (**You can not do Breakout since it will be a topic for a future studio assignment!)**
- What unique twist are you adding to make it different? (e.g., new mechanics, modernized visuals, additional features)

We are taking inspiration from Galaga. We are going to turn it into more of a bullet hell-style game with power-ups, boss fights, and an endless mode in a 2D environment.

Example: *Our game is inspired by Breakout, but instead of a paddle at the bottom, the player controls a magnet that can attract or repel the ball, adding a physics-based challenge.*

Core Gameplay

Game Loop:

Describe the basic actions the player takes and how they interact with the game.

- What is the player doing repeatedly? (e.g., jumping, shooting, dodging, collecting items)
- How does the game provide feedback? (e.g., visual effects, sound effects, UI updates)

The player moves around the screen (Using WASD or Arrow Keys) shooting enemies (by clicking or pressing space) and defeating bosses while avoiding projectiles. The point of the game is to collect as many points. The player has a timer showing them how long they have survived. There will be various bosses and enemies tied to the timer followed by animations and sound effects. When an enemy is killed they will be removed from the screen and spawn additional projectiles and have a small chance to drop a power-up. Colliding with powerups collects them and makes them active instantaneously.

Example:
The player moves a spaceship left and right to avoid asteroids and shoot enemies. Destroying enemies grants points and occasional power-ups, making the ship stronger.

Player Controls

List the controls clearly and simply.

Example:

- Keyboard (or Controller) Inputs:
  - Move: Arrow Keys / A & D
  - Jump: Spacebar
  - Attack/Shoot: Left Mouse Button / Ctrl
  - Special Ability: Shift / Right Mouse Button

Shoot is left mouse button or space

WASD/ Arrow keys =movement

Level & Progression

Game Structure:

- How many levels are in the game? (e.g., 5 levels, 10 waves, endless mode)
- Are levels handcrafted or procedurally generated?

Progression System:

- How does difficulty increase? (e.g., faster enemies, more obstacles, limited time)
- Are there power-ups or upgrades? (e.g., speed boost, double jump, new weapons)

The game will feature 1 handcrafted endless mode while the enemy spawns will be procedurally generated with bosses every couple of minutes. After a certain amount of time, enemies' stats will increase alongside projectile speed and the number of projectiles spawned by the enemies. There will be multiple power-ups available to clear the game screen of projectiles and enemies, turn immune to damage (using a shield), boost the bullet speed they have for a limited time, make movement speed faster, slow down time, and allow you to shoot multiple bullets at the same time.

Example:
Each level introduces new enemy types and increases their speed. The player can collect shield power-ups for temporary invincibility. There will be 3 levels in total and the player has to win in all 3 to finish the game

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Scoring & Win/Loss Conditions

Winning:

- How does a player complete the game? (e.g., reach the final level, defeat a boss, survive a time limit)

Losing:

- What causes a game over? (e.g., losing all lives, running out of time, missing too many targets)

Score System:

- How is progress tracked? (e.g., points, time, collected items)
- Are there multipliers, streaks, or bonuses?

Example:
The player earns points for defeating enemies. A combo multiplier increases score for consecutive hits. The game ends when the player loses all three lives.

Players earn points by defeating enemies and their score multiplier increases as time goes by, the game ends when all lives are expended and the players time and score are shown to the player prompting them to restart or leave the game.

Timeline & Milestones

Week 1: Core Mechanics & Gameplay Elements

Example:

- Set up a basic Unity project <Name (of student assigned to the task)>

- Implement player input and core mechanics (moving, jumping, shooting, etc.) <Name>
- Implement scoring system <Name>
- Create a rough UI (score display, health, etc.) <Name>

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1. Set up a unity project (Darvesh)
2. Set up a git hub and unity collab group (all)
3. Implement player movement and shooting (Sam)
4. Research enemy pathing and implement it (Cameron)
5. Find Assets (Darvesh)
6. implement game music (Darvesh)

Week 2: Core mechanics

Example:

- Design and implement levels/waves <Name>
- Add particle effects, sound effects, and other polish <Name>
- Improve UI and menus <Name>
- Package the final build <Name>

1. implement timer (Sam)
2. power-ups (Darvesh)
3. Scoring / Assign points for enemies (Darvesh)
4. Design level and Level UI (Sam)

Week 3: UI and polish

1. Menu and High Score (Darvesh)
2. Leaderboard??? (local)

Week 4: finalization

1. troubleshoot (Cameron)
2. finalize menus.(Everyone)
3. Play Testing (Everyone + Friends)

Assets

Models & Art:

Provide links or sources for sprites, models, and animations.

- Custom-made or downloaded (e.g., OpenGameArt, Kenney Assets)
- https://opengameart.org/content/starshooter
- https://opengameart.org/content/space-ship-construction-kit
- https://opengameart.org/content/space-ship-mech-construction-kit-2
- https://opengameart.org/content/lasers-and-beams
    - https://opengameart.org/content/2d-shooter-effects-alpha-version
    - https://opengameart.org/content/sci-fi-effects
    - https://opengameart.org/content/animated-particle-effects-2
    - https://opengameart.org/content/2d-explosion-animations-frame-by-frame

- https://opengameart.org/content/space-game-art-pack-extended
- https://opengameart.org/content/space-shooter-redux
- https://opengameart.org/content/space-shooter-art
- https://opengameart.org/content/modular-ships
- https://opengameart.org/content/superpowers-assets-various-2d
- https://opengameart.org/content/space-background-3
- https://opengameart.org/content/mech-icons-for-a-tactical-mech-roguelike
- https://opengameart.org/content/galagian-space-shooter-3d-prerender-ships-backgruond-ui
- https://opengameart.org/content/arcade-space-shooter-game-assets

Sound & Music:

Provide links or sources for music and sound effects.

- Royalty-free or original compositions (e.g., Freesound, BFXR, Chiptone)
- https://opengameart.org/content/rise-of-a-hero
- https://opengameart.org/content/starshooter
- https://opengameart.org/content/free-chiptune-music-package-for-game-projects-part-1

UI:

Provide links or details on icons, fonts etc.

- https://opengameart.org/content/ui-pack-sci-fi

- https://opengameart.org/content/inventory-ui
- https://opengameart.org/content/bbs-item-frames
- https://opengameart.org/content/game-ui-simple-outline-circles
- https://opengameart.org/content/galagian-space-shooter-3d-prerender-ships-backgruond-ui
- https://craftpix.net/freebies/free-space-shooter-game-gui/
- https://opengameart.org/content/free-ui-hologram-interface
- https://opengameart.org/content/arcade-space-shooter-game-assets

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